

CONSTRUCTION MANAGEMENT

Construction management or Construction project management (CPM) is the overall planning, coordination, and control a project from beginning completion. CPM is aimed at meeting a client's requirement in order to produce a functionally and financially viable project CPM is project management that applies to the construction sector. The construction industry is composed of five sectors: residential. Commercial, heavy civil, industrial, and environment. A construction manager holds the same responsibilities and completes the same processes in each sector. All that separates a construction manager in one sector from one in another is the knowledge of the construction site. This may include different types of equipment, materials, subcontractors, and possibly locations. A contractor is assigned to a construction project once the design has been completed by the architect or is still in progress. This is done by going through a bidding process with different contractors. The contractor is selected by using one of the three selection methods: low-bid selection, best –value selection. A construction manager should have the ability to handle public safety, time management, decision making, mathematics, and human resources.

FUNCTION

The functions of Construction management typically include the following:

1. Specifying project objectives and plans including delineation of scope, budgeting, scheduling, setting performance requirements, and selecting project participants.
2. Maximizing the resource efficiency through procurement of labor, materials and equipment.
3. Implementing various operations through proper coordination and control of planning, design, estimating, contracting and construction in the entire process.
4. Developing effective communications and mechanisms for resolving conflicts.

CONSTRUCTION MANAGER

Required knowledge

- Construction and Building
- Technology
- Public Safety
- Customer Service
- Human Resources
- Mathematics

Skill and Abilities

- Time management
- Negotiation
- Decision Making
- Problem Solving

The construction manager offers advice, uncolored by any conflicting interest, on matters such as:

- Optimum use of available funds;
- Control of the scope of the work;
- Project scheduling;
- Optimum use of design and construction firms' skill and talents;
- Avoidance of delays, changes and disputes;
- Enhancing project design and construction quality;
- Optimum flexibility in contracting and procurement
- Cash flow Management

Comprehensive management of every stage of the project, beginning with the original concept and project definition, yields the greatest possible benefit to owners from Construction Management. As time progress beyond the pre-design phase, the CM's ability to effect cost savings diminishes. The Agency CM can represent the owner by helping to select the design team as well as the construction team and manage the design preventing scope creep, helping the owners stay within a pre-determined budget by performing Value Engineering Cost/ Benefit Analysis and Best Value Comparison. The software application field of construction collaboration technology has been developed to apply information technology to the construction management process.

CM at – risk

CM at – risk is a delivery method which entails a commitment by the construction manager to deliver the project within a Guaranteed Maximum Price (GMP), in most cases. The construction manager acts as consultant to the owner in the development and design phases, (often referred to as “Preconstruction Services”), but as the equivalent of a general contractor during the construction phase. When a construction manager is bound to a GMP, the most fundamental character of the relationship is changed. In addition to acting in the owner's interest, the construction manager must manage and control construction costs to not exceed the GMP, which would be a financial hit to the CM Company.

Planning and Scheduling

Project management methodology:

- Work breakdown structure
- Project network of activities
 - Critical path method (CPM)
 - Resource management
 - Resource leveling

Architecture – Engineer

- Work inspection
- Change order
- Review payment
- Materials and samples
- Shop drawings
- 3d image

Agency CM

Construction Cost Management is a fee-based services in which the Construction Manager (C.M) is responsible exclusively to the owner and acts it the owner's interests at every stage of the project. The construction manager offers advice, uncolored by any conflicting interest, on matters such as:

- Optimum use of available funds;
- Control of the scope of the work;
- Project scheduling;
- Optimum use of design and construction firms' skills and talents;
- Avoidance of delays, changes and disputes;
- Enhancing project design and construction quality;

CONSTRUCTION MANAGEMENT

Accounting & Operational for Contractors, Building & Civil

Duties

A construction project manager sets up the estimates, the budgets and the construction timetable for the client and develops the construction strategy. He selects the subcontractors and workers, and provides required explanations for the builders and other professional associated with the project, coordinating and collaborating with the architects, engineers and specialists. The construction manager also ensures the construction project complies with all building codes and any other legal or regulatory requirements. When delays or problems occur- as they always do- the construction manager is the project's first responder, ready to make the changes required to move the project past the issue.

OBTAINING THE PROJECT

Bids

A bid is given to the owner by construction manager that are willing to complete their construction project. A bid tells the owner how much money they should expect to pay the construction management company in order for the complete the project.

- **Open Bid:** An open bid is used for public projects. Any and all contractors are allowed to submit their bid due to public advertising.
- **Closed Bid:** A closed bid is used for private projects. A selection of contractors are sent an invitation for bid so only they can submit a bid for the specified project.

Selection Methods

- **Low- bid selection:** This selection focuses on the price of a project. Multiple construction management companies submit a bid to the owner that is the lowest amount they are willing to do the job for. Then the owner usually chooses the company with the lowest bid to complete the job for them.
- **Best – Value selection:** This selection focuses on both the price and qualifications of the contractors submitting bids. This means that the owner choose the contractor with the best prices and the best qualifications. The owner decides by using a request for proposal (RFP), which provides the owner with the contractors exact form of scheduling and budgeting that the contractor expects to use for the project.
- **Qualifications – Based selection:** This selection is used when the owner decides to choose the contractor only on the basis of their qualifications. The owner then uses a request for qualifications (RFQ), which provides the owner with the contractor's experience, management plans, project organization, and budget and schedule performance. The owner may also ask for safety records and individual credentials of their members.

Payment Contracts

- **Lump – sum:** This is the most common type of contract. The construction manager and the owner agree on the overall cost of the construction project and the owner is responsible for paying that amount whether the construction project exceeds or falls below the agreed price of payment.
- **Cost – Plus –fee:** This contract provides payment for the contractor including the total cost of the project as well as a fixed fee or percentage of the total cost. This contract is beneficial to the contractor since any additional costs will be for paid even though they were unexpected for the owner.
- **Guaranteed Maximum Price:** This contract is the same as the cost – plus – free contract although there is a set price that the overall cost and fee do not go above.
- **Unit – Price:** This contract is used when the cost cannot be determined ahead of time. The owner provides materials with a specific unit price to limit spending.

The Five Types of Construction

- **Residential:** Residential housing is for society including individual homes, apartments, condominium, and townhouses.
- **Commercial:** This refers to construction dealing with the needs of commerce, trade, and government. Some examples may be school, banks, hospitals, theaters, and government buildings.
- **Heavy Civil:** This type of construction greatly impacts society since it deals with transportation. Some examples are roads, bridge, railroad, dams, tunnels, and airports.
- **Industrial:** Industrial is connected will buildings that are used for different types of production. Some examples are chemical plants, steel mills, oil refineries, manufacturing plants, and pipelines.
- **Environmental:** Environmental used to be part of heavy civil, but it now has its own section since it deals with projects that improve the environment. Some examples are sanitary sewers, waste management, and clean water.

Construction Management Jobs

- Planning Engineer
- Project Coordinator
- Field Engineer
- Office Engineer
- Quantity Surveyor
- Project Engineer
- Area Superintendent
- Project Manager
- Estimator
- Lead Estimator
- Senior Estimator
- Chief Estimator

Obtaining the Project

- **Field Engineer:** A field engineer is considered an entry-level position and is responsible for paperwork.
During the pre-construction stage, a site investigation must take place. A site investigation takes place to discover if any steps need to be implemented on the job site. This is in order to get the site ready before the actual construction begins. This also includes any unforeseen conditions such as historical artifacts or environment problems. A soil test must be done to determine if the soil is in good condition to be built upon.

Procurement

The procurement stage is when labor, materials and equipment needed to complete the project are purchased. This can be done by the general contractor if the company does all their own construction work. If the contractor does not do their own work, they obtain it through subcontractors.

Subcontractors are contractors who specialize in one particular aspect of the construction work such as concrete, melding, glass, or carpentry. Subcontractors are hired the same way a general contractor would be, which is through the bidding process. Purchase orders are also part of the procurement stage.

- **Purchase Order:** A purchase order is used in various types of businesses. In this case, a purchase order is an agreement between a buyer and seller that the products purchased meet the required specification for the agreed price.

Construction

The construction stage begins with a pre-construction meeting brought together by the superintendent. The pre-construction meeting is meant to make decisions dealing with work hours, material storage, quality control, and site access. The next step is to move everything onto the construction site and set it all up.

At this stage, construction monitoring and supervision is of great importance to ensure that a project is completed on time and on budget, while meeting all relevant regulations and quality standards.

Post – Construction

Once the construction has been completed there are specific steps that must be taken to prepare the building for occupancy.

- **Project Punch out:** A project punch out means that the project must be looked at for any issues before it is considered completely finished. Issues may include replacing a cracked tile on the floor or changing the color of paint. A list is created containing these issues and it is known as a punch list.
- **Substantial Completion:** The architect for the project determines if the building meets every requirement and issues a certificate of substantial completion.
- **Guaranteed Maximum Price:** This contract is the same as the cost – plus- fee contract although there is a set price that the overall cost and fee do not go above.
- **Unit – Price:** This contract is used when the cost cannot be determined ahead of time. The owner provides materials with a specific unit price to limit spending.

Project Stages

Design

The design stage contains a lot of steps: programming and feasibility, schematic design, design development, and contract documents. It is the responsibility of the design team to ensure that the design meets all building codes and regulations. It is during the design stage that the bidding process takes place.

- **Programming and Feasibility:** The needs, goals, and objectives must be determined for the building. Decisions must be made on the building size, number of rooms, how the space will be used, and who will be using the space. This must all be considered to begin the actual designing of the building.
- **Schematic Design:** Schematic designs are sketches used to identify spaces, shape, and patterns. Materials, sizes, colors, and textures must be considered in the sketches.
- **Design Development (DD):** This step requires research and investigation into what materials and equipment will be used as well as their cost.
- **Contract Documents (CDs):** Contract documents are the final drawings and specifications of the construction project. They are used by contractors to determine their bid while builders use them for the construction process. Contract documents can also be called working drawing.

Pre – Construction

The pre-construction stage begins when the owner gives a notice to proceed to the contractor they have chosen through the bidding process. A notice to proceed is when the owner gives permission to the contractor to begin their work on the project. The first step is to assign the project team which includes the project manager (PM), contract administrator, superintendent, and field engineer

- **Project manager:** The project manager is in charge of the project team.
- **Contract Administrator:** The contract administrator assists the project manager as well as the superintendent with the details of the construction contract.

- **Superintend:** It is the superintendent's job to make sure everything is on schedule including flow of material, deliveries, and equipment. They are also in charge of coordinating on-site construction activities.
- **Field Engineer:** A field engineer is considered an entry-level position and is responsible for paperwork.

During the pre-construction stage, a site investigation must take place. A site investigation takes place to discover if any steps need to be implemented on the job site. This is order to get the site.

- **Substantial Completion:** The architect for the project determines if the building meets every requirement and issues a certificate of substantial completion. This certificate announces the official completion of the project.
- **Final Inspection:** A final inspection is done by the building official once the certificate of substantial completion has been issued.
- **Certificate of Occupancy:** A certificate of occupancy is issued by the building official which informs the owner that it is now safe to occupy. This is issued by the building official after the final inspection.
- **Final Documentation:** This provides information on the building to the owner for future references. This includes warranties, operation manuals, inspection and testing reports, and record drawings.
- **Final Completion:** Final completion occurs when all required paperwork and documentation is completed, including payments to the contractor.

Owner Occupancy

Once the owner moves into the building, a warranty period begins. This is to ensure that all materials, equipment, and quality meet the expectations of the owner that are included within the contract.

Issues Resulting from Construction

Noise Control

Dust and Mud

Environmental Protections

- **Storm water Pollution:** As a result of construction, the soil is displaced from its original location which can possibly cause environmental problems in the future. Runoff can occur during storms which can possibly transfer harmful pollutants through the soil to rivers, lakes, wetlands, and coastal waters.
- **Endangered Species:** If endangered species have been found on the construction site, the site must be shut down for some time. The construction site must be shut down for as long as it takes for authorities to make a decision on the situation. Once the situation has been assessed, the contractor makes the appropriate accommodations to not disturb the species.
- **Vegetation:** There may often be particular trees or other vegetation that must be protected on the job site. This may require fences or security tape to warn builders that they must not be harmed.
- **Wetlands:** The contractor must make accommodations so that erosion and water flow are not affected by construction. Any liquid spills must be maintained due to contaminants that may enter the wetland.
- **Historical or Cultural artifacts:** Artifacts may include arrowheads, pottery shards, and bone. All work comes to a halt if any artifacts are found and will not resume until they can be properly examined and removal from the area.

Construction Activity Documentation

Project Meetings

Project meetings take place at scheduled intervals to discuss the progress on the construction site and any concerns or issues. The discussion and any decisions made at the meeting must be documented.

Diaries, Logs, and Daily Field Reports

Diaries, logs, and daily field reports track of the activities on a job site each day.

- **Diaries:** each member of the project team is expected to keep a project diary. The diary contains summaries of the day's events in the member's own words. They are used to keep track of any daily work activity, conversations, observations, or any other relevant information regarding the construction activities. Diaries can be referred to when disputes arise and a diary happens to contain information connected with the disagreement. Diaries that are handwritten can be used as evidence in court.
- **Logs:** Logs keep track of the regular activities on the job site such as phone logs, delivery logs, and RFI (Request for Information) logs.
- **Daily Field Reports:** Daily field reports are a more formal way of recording information on the job site. They contain information that includes the day's activities, temperature and weather conditions, delivered equipment or materials, visitors on the site, and equipment used that day.

Labor

Visual Records

Web Cameras

Resolving Disputes

- **Mediation:** Mediation uses a third party mediator to resolve any disputes. The mediator helps both disputing parties to come to a mutual agreement. This process ensures that no attorneys become involved in the dispute and is less time-consuming.
- **Minitrial:** A minitrial takes more time and money than mediation. The minitrial takes place in an informal setting and involves some type of advisor or attorney that must be paid. The disputing parties may come to an agreement or the third party advisor may offer their advice. The agreement is nonbinding and can be broken.
- **Arbitration:** Arbitration is the most costly and time-consuming way to resolve a dispute. Each party is represented by an attorney while witnesses and evidence are presented. Once all information is provided on the issue, the arbitrator makes a ruling which provides the decision. The arbitrator provides the final decision on what must be done and it is a binding agreement between each of the disputing parties.

SUMMARY OF CONSTRUCTION MANAGEMENT